Growing crops

9B Plant growth · Lesson 5 of 7

Date _____ Class _____

Today I can

- ☐ I can explain how fertilisers, pesticides, greenhouses and machines increase crop yield, and how pests and population growth change the food supply.
- ☐ I can describe how new varieties are made by cross-breeding and selective breeding, and interpret the results of a fertiliser investigation.

What happens today

1. Do Now: more from the same field
2. Learn: Fertilisers and the best conditions · A fertiliser investigation · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 28 to 31 · Workbook pages 20 to 22

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
yield	the amount of useful product a farmer gets from a crop
fertiliser	a substance added to soil that contains mineral salts to help plants grow
manure	animal waste used as a natural fertiliser
decomposer	a microorganism that breaks down dead material and waste
pesticide	a substance that kills pests (organisms that damage crops)
insecticide	a pesticide that kills insects
fungicide	a pesticide that kills fungi that cause plant diseases
herbicide	a pesticide that kills weeds
weed	a plant growing where it is not wanted, competing with the crop
variety	a group of plants bred to have certain characteristics
selective breeding	breeding only from the plants with the wanted characteristics, generation after generation

Quick Review

more from the same field

1 What does the yield of a crop mean?

2 Name three things a crop plant needs that a farmer can provide.

3 Suggest why a farmer might remove a hedge from the edge of a field.

Core 1 of 4

Everyone does these. Answer on the card or under it.

1 Match each type of pesticide to what it kills. Draw a line.

insecticide	weeds with broad leaves, but not narrow-leaved crops such as wheat
fungicide	insect pests such as aphids
herbicide	all weeds (plants competing with the crop)
selective herbicide	fungi that cause plant diseases such as rust

Core 2 of 4

Everyone does these. Answer on the card or under it.

2 Students investigated fertiliser. Each group grew 20 bean plants in identical pots, one plant per pot, and added a different number of drops of the same liquid fertiliser. After a week they calculated the mean increase in height.

a) Plot the results as a bar chart on the grid. Label both axes.

Group	Drops of liquid fertiliser	Mean increase in height (cm)
1	2	4.6
2	4	6.3
3	6	7.8
4	8	7.2

Core 3 of 4

Everyone does these. Answer on the card or under it.

2 *continued*

b) Describe the trend in the results between groups 1, 2 and 3.

c) What advice would you give a farmer about how much fertiliser to use, based on the result for group 4? Give a reason.

Core 4 of 4

Everyone does these. Answer on the card or under it.

3 Complete the sentences. Use the word bank.

Fertilisers contain _____ _____ that help plants grow. Farmers use cheap _____ fertilisers and natural ones such as _____ (animal waste).

Microorganisms called _____ break down manure and release the mineral salts, but this takes a long time.

Challenge 1 of 2

When Core is finished. Full sentences.

C1 An apple breeder has a yellow apple variety that is not sweet and a red apple variety that is sweet. Describe how the breeder could produce a new yellow, sweet variety. Use the words cross-breeding, select and generations.

Challenge 2 of 2

When Core is finished. Full sentences.

C2 Farms in the UK today produce 35% more food per square metre than they did in 1973. Explain three different ways farmers have achieved this.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What do fertilisers contain that helps plants grow?

2 What do herbicides kill, and why does this help the crop?

3 Describe one step in selective breeding.

Today I can: Goal 1 ☐ Goal 2 ☐